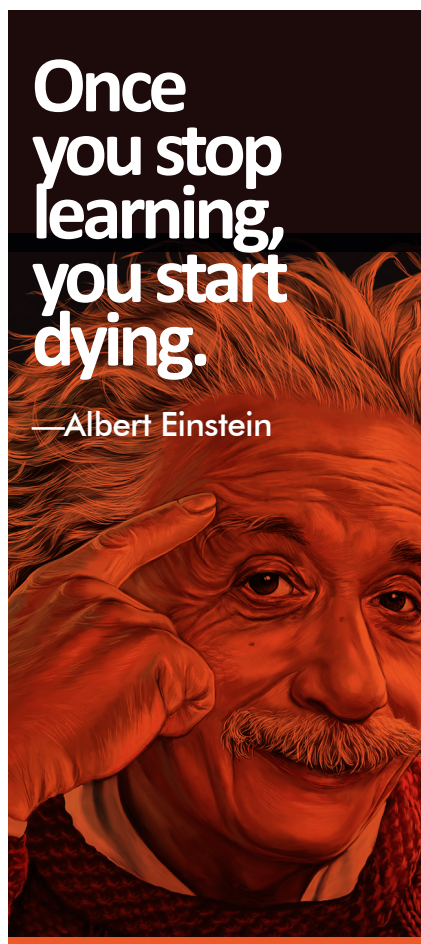




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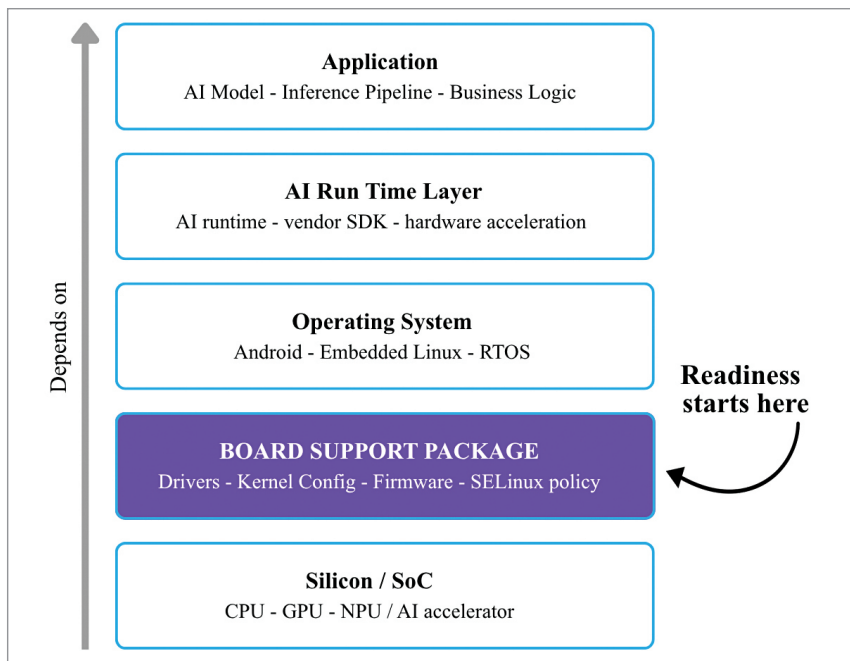


Fig. 2: Where BSP readiness sits beneath every workload

validation—confirming that cameras, sensors, and other inputs and outputs keep behaving correctly while the NPU is under load—matters just as much as the model's own accuracy score.

Skipping these checks can leave a system that worked perfectly in the lab missing detections on the factory floor within its first week of continuous use.

Security and updates cannot wait

An edge AI device that works well today still has to work safely a year from now, which raises two questions that are easy to defer and expensive to ignore.

The first is security. Android-based industrial devices benefit from SELinux (security-enhanced Linux) policy hardening, a set of mandatory access controls that limit what any single process, including the AI inference pipeline, is allowed to touch on the system. Getting this wrong either leaves the device exposed or breaks legitimate functionality when policies are configured too restrictively; getting it right takes deliberate tuning, not a default

configuration copied from a reference build.

The second is lifecycle management. Models drift, vulnerabilities get discovered, and firmware needs patching. None of that is possible without a reliable over-the-air (OTA) update pipeline and a device management approach that can be trusted across an entire fleet of deployed units, not just the one test unit sitting on an engineer's desk.

Readiness is a team sport

A good model is necessary, but never sufficient. Systems that make it from validation to the factory floor involve embedded platform engineers from the first week, rather than handing them a finished model and asking them to make it run.

That means treating BSP readiness, hardware-software alignment, real-world validation, and lifecycle planning as part of the AI system itself, rather than problems to solve after the fact. Industrial edge AI rarely fails because the algorithms are wrong. More often, it fails because the platform underneath them was not considered until it was too late. **EFY**